BIG DATA ANALYTICS

SMART METER ELECTRICITY CONSUMPTION

Distributed Processing of High-Frequency Electricity Data from Mathura and Bareilly

Case Study : 1 | Presented By: Harshita(160124733216) | Hridya(160124733215) | Namratha (160124733218) | Priyanka (160124733220)

DATASET OVERVIEW

Property	Details
Dataset	Smart Meter Data: Mathura & Bareilly
Location	Mathura & Bareilly, Uttar Pradesh
Smart Meter	Nearly 50
Time Period	May 2019 - October 2021
Sampling Interval	3 Minutes
Files	8 Minutes

6,627,369
Total reading(rows)

46
Unique Smart meters

308.4 MB
Size of the CSV

Schema:

- x_timestamp
- t_kWh
- z_Avg Voltage(V)
- z_Avg Current(A)
- y_Freq (Hz)
- meterID

DATA FLOW

1

STEP 1
Data Collection

2

STEP 2
Upload to HDFS

3

STEP 3
Hdfs Blocks

4

STEP 4
Map reduce jobs(includes MapReduce Joins)

5

STEP 5
Mapper

6

STEP 6
Shuffle and sort

7

STEP 7
Reducer

8

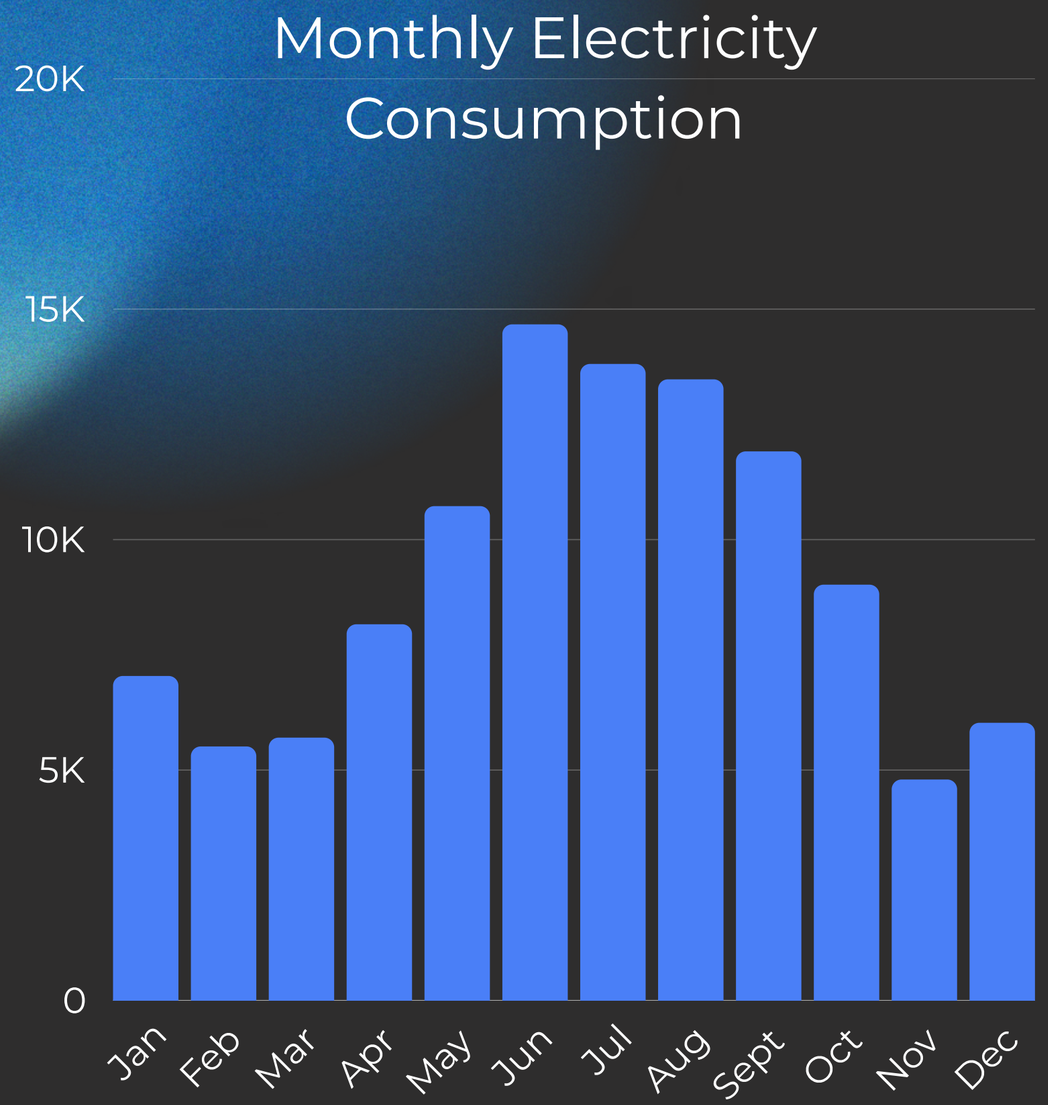
STEP 2
Output

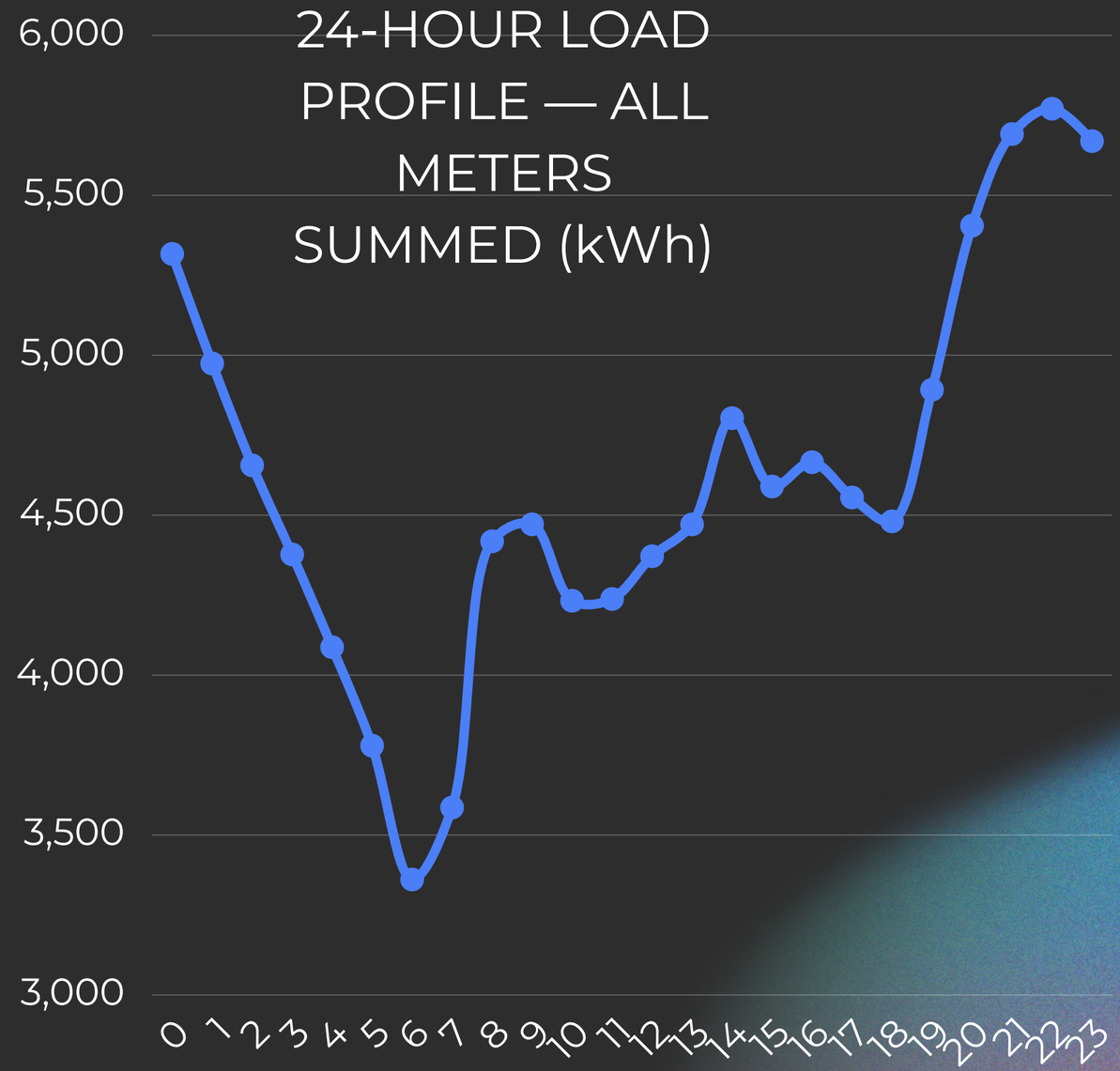
9

STEP 3
Analysis and Data Visualisation

CSV → HDFS → MAP → SHUFFLE → REDUCE → RESULT

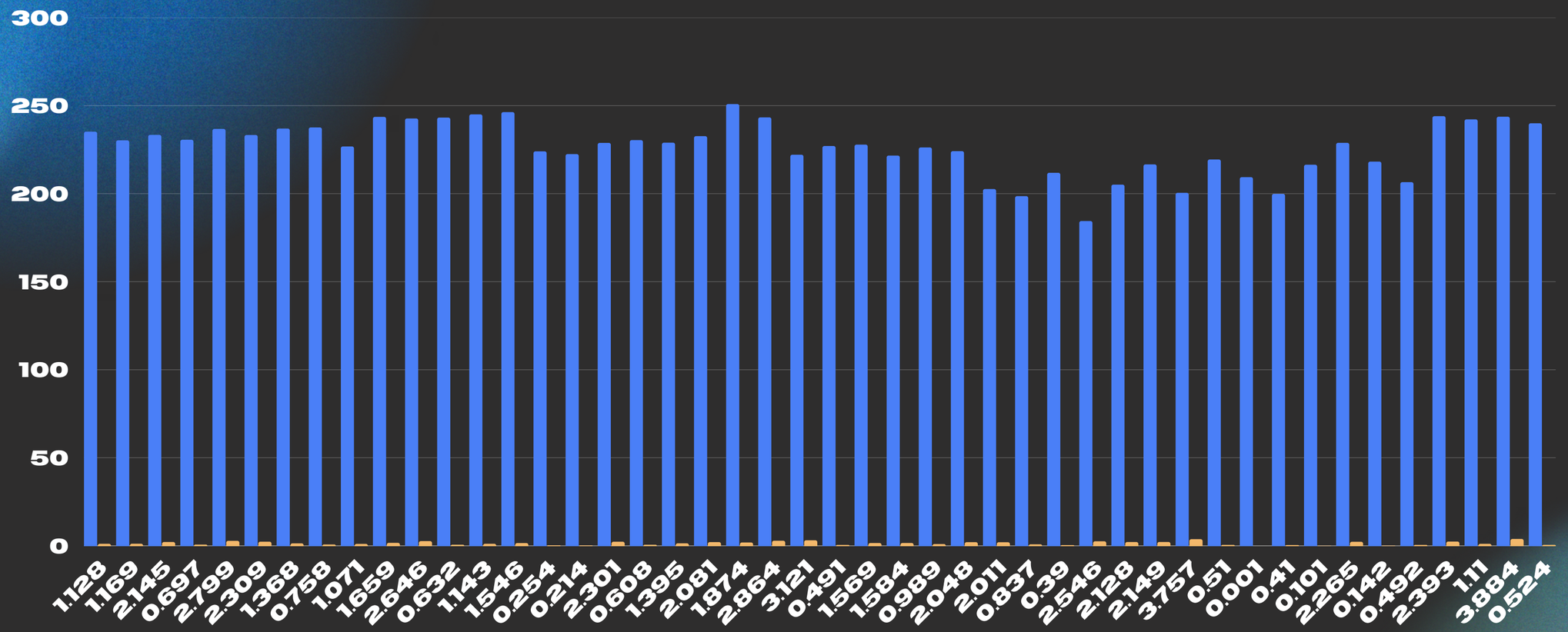
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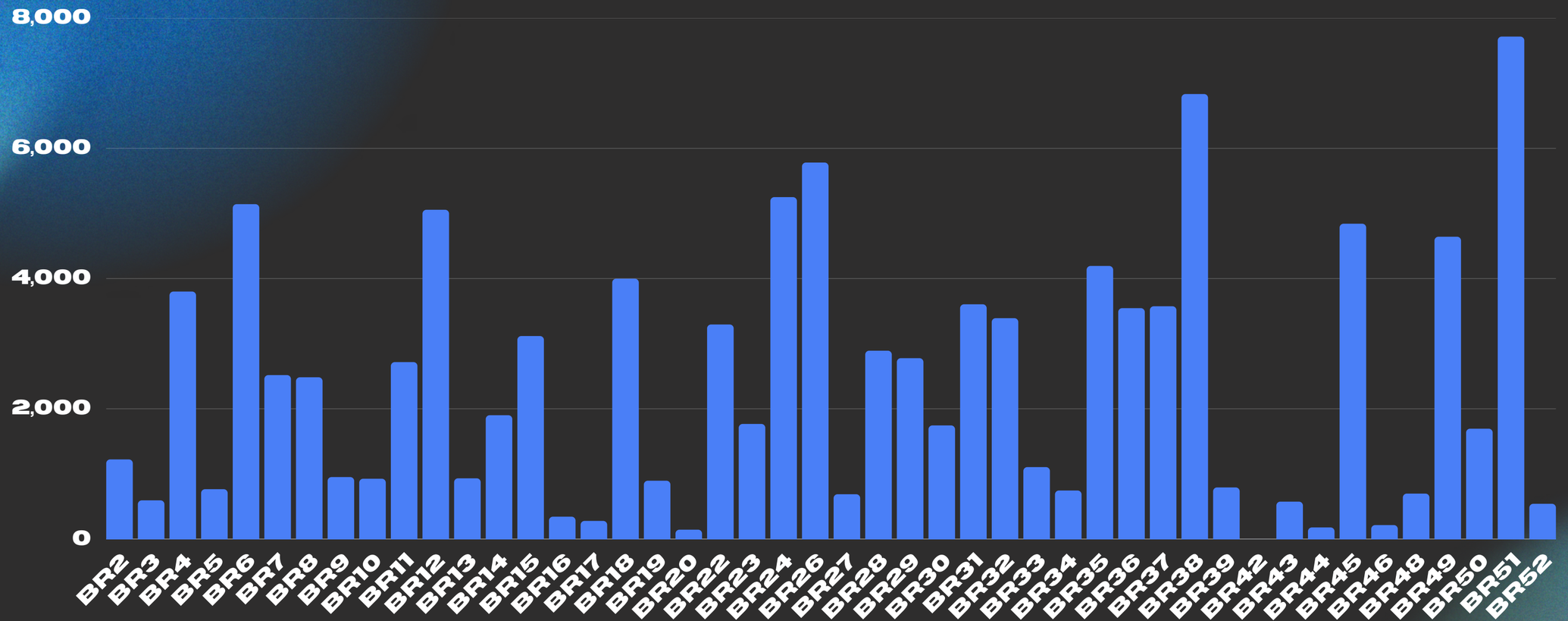
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Voltage and current distribution

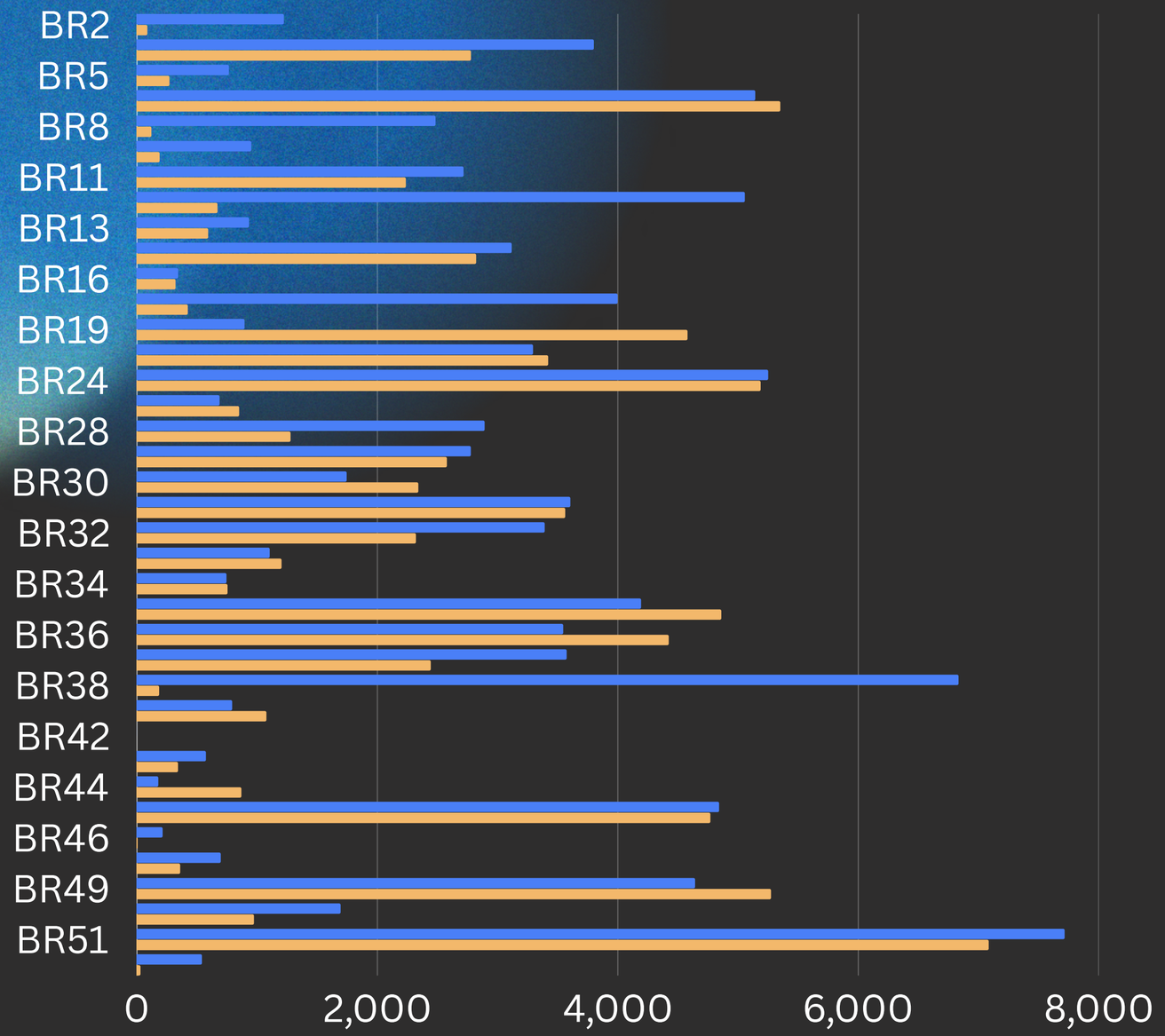


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Meter-wise Consumption



MAPREDUCE JOIN OUTPUT



- Some meters showed a large increase in 2021, such as BR19 and BR44.
- Some meters showed a large decrease in 2021, such as BR38, BR08 and BR12.
- Electricity usage can vary significantly from one year to another for individual meters.
- The Map-Side Join and Reduce-Side Join give exactly the same results, showing that both methods successfully joined the 2020 and 2021 data correctly.

INTERPRETATION & IMPLICATIONS

Seasonal load follows heat

Consumption triples from the November low to the June peak (4,796 → 14,669 kWh), tracking the pre-monsoon summer cooling season rather than any fixed monthly pattern.

Low voltage issue

9 meters run below the safe limit of 216V, and only 4 run above 243V. The worst is BR35 at just 184V. So the issue here is voltage drop, not over-voltage.

Demand is highest in the evening

Demand is lowest at 6 AM (3,361 kWh) and highest at 10 PM (5,771 kWh). Load builds up all evening and stays high past midnight. This is a normal household pattern, not an industrial one.

Consumption is concentrated

The top 5 of 46 meters account for roughly 28% of total annual energy (30,727 of 110,862 kWh) — a small share of households drive a disproportionate share of load.

A few meters use most of the power

The top 5 meters out of 46 use 28% of all electricity. The top 10 use 48%. So half the total load comes from one-fifth of the connections.

The MapReduce Join reveals key details

Total usage fell 21% from 2020 to 2021, but most meters stayed steady. The fall came from a few meters that dropped over 90% (BR38, BR02, BR08). Those are disconnections or missing data, which need to be checked.

CONCLUSION

Infrastructure

A 308.4 MB CSV file was uploaded to HDFS, split into 3 blocks, and processed by 1 master + 3 worker nodes. Each worker ran a Map task on its own local block (data locality), a combiner compressed the data before shuffling, and 3 Reduce tasks combined everything into the final result.

Consumption Findings

110,862 kWh total across 46 meters. Usage triples from November to June (heat-driven). Peak load happens at night, around 10 PM. Top 10 meters (22% of connections) use 48% of all electricity.

Join Findings

Map-side and reduce-side joins gave identical results, confirming correctness. 2020–2021 comparison shows a 21% overall drop, but it's driven by a few meters collapsing (BR38, BR02, BR08) rather than real demand reduction — most meters stayed steady, and some even rose sharply (BR19, BR44).

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THANK YOU

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